



Summer Semester July 2024
ME1006-1 – MANUFACTURING PROCESSES
(ME)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- Which of the following is 'not' an allowance given to the pattern for casting?
A) Shrinkage
B) Draft
C) Hole
D) Machining
- What is the purpose of the sprue in sand casting?
A) To melt the metal.
B) To create the mold cavity.
C) To remove the casting from the mold.
D) To channel the metal from the pouring basin to the gating system.
- What is the purpose of the core in sand casting?
A) To create the mold cavity.
B) To remove the casting from the mold.
C) To melt the metal.
D) To create internal features in the casting.
- What is the purpose of the parting line in sand casting?
A) To separate the cope and drag halves of the mold.
B) To create the mold cavity.
C) To remove the casting from the mold.
D) To pour the metal into the mold.
- Which of the following is a principal binder used in carbon dioxide moulding?
A) Carbon
B) Carbon Dioxide
C) Sodium
D) Sodium silicate
- Which type of sand is used in shell moulding?
A) Black sand
B) Wet and fine sand
C) Dry and fine sand
D) Green sand
- Another name for Gravity Die Casting is.
A) Centrifugal casting
B) Permanent mould casting
C) Precision investment casting
D) Hot chamber process
- Pattern used in shell moulding is normally made up of _____
A) Wax
B) Metal
C) Wood
D) Plastic
- Which of the following components is mainly manufactured by performing metal forging?
A) Piston
B) Engine block
C) Connecting rod
D) Crankcase
- Which of the following metal forming processes performs squeezing out of material through a hole?
A) Forging
B) Rolling
C) Drawing
D) Extrusion
- Which of the following components are manufactured by the sheet metal forming process?
A) Engine blocks
B) Connecting rods
C) Electric wires
D) Car bodies

12. Which of the following is not a type of rolling mill?
 A) Two-high rolling mill B) Cluster rolling mill
 C) Separation rolling mill D) Tandem rolling mill
13. How is the die used in injection molding cooled?
 A) Oil B) Air
 C) Water D) Contact with cold surface
14. In blow molding, to inflate soft plastic, which medium is used?
 A) Air B) Water
 C) Oil D) Alcohol
15. Glasses are used to store acids because they are:
 A) Hard B) Brittle
 C) Chemical stable D) Refractory
16. In jiggering process, the starting material is
 A) Slip B) Gob
 C) Slug D) Powder
17. Which of the following is not a type of arc welding?
 A) Plasma B) Electro-slag
 C) Submerged D) Air-acetylene
18. Which of the following is not a type of resistance welding?
 A) Seam B) Projection
 C) Electro-slag D) Spot
19. Thermite contains which metal?
 A) Zinc B) Copper
 C) Aluminium D) Manganese
20. Which of the following types of fuel gas is commonly used in gas welding?
 A) Biogas B) Coal gas
 C) Acetylene D) Methane

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1.	a) Write a note on pattern materials.	8	L*1	1	1
	b) Write a note on any four casting defects. Mention the reasons why they occur and the remedies.	8	L2	1	1
2.	a) Explain with sketch the working of jolt squeeze machine and sand slinger.	8	L2	1	1
	b) Explain with sketch the hot and cold pressure die casting processes.	8	L2	2	1
3.	a) With a schematic representation, illustrate the features of i) Loose piece pattern ii) Gated pattern	8	L2	1	1
	b) Write a note on following special casting techniques i) Evaporative pattern casting ii) Squeeze casting	8	L1	2	1
Unit - II					
4.	a) Write a note on following in the context of forging processes i) Upsetting ii) Drawing down iii) Bending	8	L1	3	1
	b) Explain the following processes of producing ceramic products: i) Slip casting method ii) Jiggering method	8	L2	4	1
5.	a) Compare with sketch, the working of combination and progressive dies.	8	L2	3	1
	b) Explain with sketch rotomolding and thermoforming processes.	8	L2	4	1

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|----|----|--|---|----|---|---|
| 6. | a) | Explain Hydrostatic extrusion and Wire drawing process. | 8 | L2 | 3 | 1 |
| | b) | Explain with sketch, blown film extrusion method of plastic processing.. | 8 | L2 | 4 | 1 |

Unit – III

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|----|-----|--|---|----|---|---|
| 7. | a) | With a neat sketch describe the principle of Thermit welding and Friction welding techniques. | 8 | L2 | 5 | 1 |
| | b) | Describe the principle of Gas welding mentioning the chemistry of the flame formation.
Write the nature of following types of flames obtained from gas welding torch, | | | | |
| | i) | Carburizing flame | 8 | L2 | 5 | 1 |
| | ii) | Oxidizing flame | | | | |
| 8. | a) | Illustrate the process of submerged arc welding process. Mention its disadvantages | 8 | L2 | 5 | 1 |
| | b) | With a schematic representation, explain the construction and working of a Plasma arc welding process and mention its advantages. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
